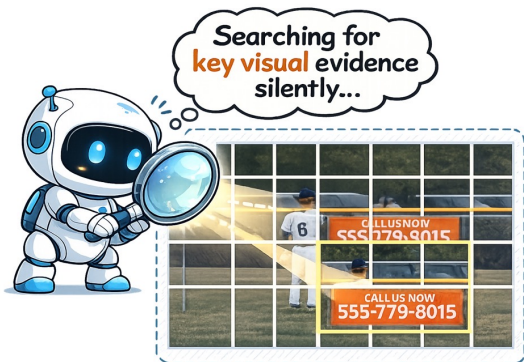


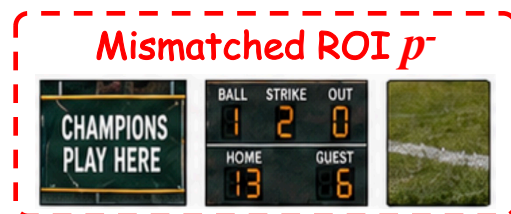
RealVR trains latent tokens with differential readout credit and visual evidence supervision



RealVR additions · training only
Inference: original LVR procedure



Align with p^+



Separate from p^-

Visual contrastive loss

$$\ell_{ev} = \sum_t \text{sg}(w_t) [m_{ev} - g_t]_+$$

Added objective: $\lambda_{ev} \hat{L}_{ev}$

Training only

trains latent generation

z_t

weights

Answer contrast

Training only

$$\gamma_t = [r_t^+ - r_t^-]_+$$

Detached weights w_t

Correct answer
8015

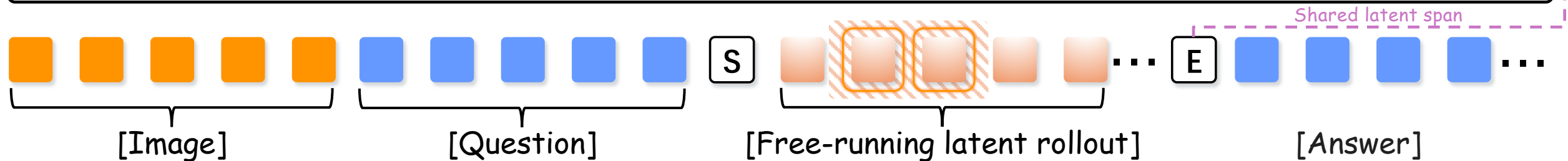
Wrong answer
7915

Teacher-forced
after shared span

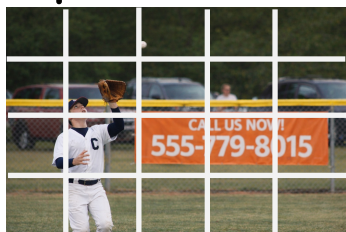
Decoder attention

Readouts r_t^+, r_t^-

Multi-Modal Large Language Model



Input



What are the last four
digits of the phone number
on the banner?

Token legend

Image
Embedding

Text
Embedding

Latent
Token

Higher-credit
Latent Token

Special Token
<lvr_start>

Special Token
<lvr_end>